



JAIPUR NATIONAL UNIVERSITY

Semester/Year End (Main/Back) Examinations, May-2025

Name of School:	School of Engineering and Technology
Name of Programme / Course:	B.Tech. (Computer Science) and CSE (Hons.) in AI and ML
Semester/Year:	Semester-IV
Paper Code:	BTECSCO401T22
Title of Paper:	Optimization Techniques
Time: 3 Hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

This question paper consists of two sections, Section-A and Section-B. Section-A consists of seven questions and Section-B consists of six questions. Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question.

Question Number	SECTION-A	Marks Allotted
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Q.1 Find the optimal solution (maxima-minima) of the function $F(x)=4x^3-18x^2+27x-7$ 4

Q.2 Job (i) on machine (j), how should the jobs be assigned to the various machines so that the cost is minimum 4

	m_1	m_2	m_3	m_4
J_1	12	30	21	15
J_2	18	33	9	31
J_3	44	25	24	21
J_4	23	30	28	14

Q.3 Write the solution of 2×2 game without saddle point 4

Q.4 Define the Dominance Rule of the game theory . 4

Q.5 Find the solution the following problem using lagranges function 4

minimize $f(x,y)=Kx^{-2}y^{-1}$

subject $g(x,y)=x^2+y^2-a^2=0$

Q.6 Find the extreme point of the function 4

$F(x,y)=x_1^3+x_2^3+2x_1^2+4x_2^2+10$

Q.7 Solve the graphically 4

$Z=6x_1+15x_2$

subject $5x_1+3x_2\leq 15$

$2x_1+5x_2\leq 10,$

$x_1, x_2 \geq 0$

SECTION-B

Q.1 Find the minimum cost and path by North-West corner Rule of T.P. 10

	W_1	W_2	W_3	W_4	W_5	supply
F_1	7	6	4	5	9	40
F_2	8	5	6	7	8	30
F_3	6	8	9	6	5	20
F_4	5	7	7	8	6	10
Demand	30	30	15	20	5	100

Q.2 Find the optimal solution of the transportation problem by MODI method 10

	D_1	D_2	D_3	D_4	supply
O_1	3	1	7	4	250
O_2	2	6	5	9	350
O_3	8	9	3	2	400
Demand	200	300	350	150	1000

Q.3 Solve the transportation problem by using Least cost method 10

	D ₁	D ₂	D ₃	D ₄	supply
O ₁	6	4	1	5	14
O ₂	8	9	2	7	16
O ₃	4	3	6	2	5
Demand	6	10	15	4	35

Q.4 A Project schedule has the following characteristics 10

Activity	1-2	1-3	2-4	3-4	3-5	4-9	5-6	5-7	6-8	7-8	8-10	9-10
time	4	1	1	1	6	5	4	8	1	2	5	7

- (i) Construct network diagram
- (ii) Compute the earliest event time and Latest event time
- (iii) Determine critical path and total project duration (day)

Q.5 Assuming that the expected time are normally distribution 10
find the probability of meeting the schedule date as given for network

Activity	Optimistic a	most likely b	positive time c
1-2	2	5	14
1-3	9	12	15
2-4	5	14	17
3-4	2	5	8
3-5	8	17	20
4-5	6	6	12

Q.6 Construct a network for the project whose activities and their precedence relationships are given below 10

Activity	A	B	C	D	E	F	G	H	I	J	K
precedence	-	A	A	C	B,C	D,E	E	G	D,F	I,H	J